First Duplicate Value

AlgoExpert · Medium · Arrays

The problem

- Values are in [1, n]; return the value whose second occurrence comes first, else -1.
- Mark visited values by negating the slot at index |v|-1 (mutates input).

Intuition

- Values are between one and n, so each value can mark its own index by flipping that slot negative.

Approach

- 1. For each value, look at index abs value minus one
- 2. If that slot is already negative, we've seen it: return it
- 3. Otherwise flip it negative

Java solution

```java
public class FirstDuplicateValue {
    public static int firstDuplicateValue(int[] a) {
        for (int x : a) {
            int v = Math.abs(x);
            if (a[v - 1] < 0) return v;
            a[v - 1] = -a[v - 1];
        }
        return -1;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Recap

- Values are between one and n, so each value can mark its own index by flipping that slot negative.
- Complexity: O(n)